

**Day 04 -
Mathematical
Preliminaries**



$$\theta = \tan^{-1} \left(\frac{F_y}{F_x} \right)$$

Announcements

- Homework 1 is due this Friday
- Homework 2 is posted now
- Help sessions start this week
 - DC Friday at 2-4pm (1248 BPS)
- Mihir (ULA) will host additional help hours soon

Seminars this week (Wednesday)

WEDNESDAY, September 3, 2025

Astronomy Seminar, 1:30 pm, 1400 BPS, In Person and Zoom, Host~

Speaker: Group Introductions

Title:

Zoom Link: [https://msu.zoom.us/j/887295421?
pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09](https://msu.zoom.us/j/887295421?pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09)

Meeting ID: 887-295-421

Passcode: 002454

Seminars this week (Wednesday)

WEDNESDAY, September 3, 2025

Nuclear Science Seminar, 3:30Pm., FRIB 1300 lab in person and online via Zoom

Speaker: Mark Spieker, Florida State University

Title: Experimental studies of the pygmy dipole resonance

Zoom Link: [https://msu.zoom.us/j/95277003505?](https://msu.zoom.us/j/95277003505?pwd=hTILu1oLqmhTCU7jIVKFTIXXZBmuGb.1)

[pwd=hTILu1oLqmhTCU7jIVKFTIXXZBmuGb.1](https://msu.zoom.us/j/95277003505?pwd=hTILu1oLqmhTCU7jIVKFTIXXZBmuGb.1)

Meeting ID: 952 7700 3505

Passcode: 404830

Goals for this week

Be able to answer the following questions.

- What are the essential physics models for single particles?
- How do we setup problems in classical mechanics?
- What mathematics do we need to get started?
- How do we solve the equations of motion?

Reminders from Day 03

- In a Newtonian world, we start from a vector description of motion
- Differential equations are mathematical models that describe the motion of particles
- We can use different methods to solve these differential equations

i-Clicker: <https://join.iclicker.com/QTEC>



Projectile Motion

$$\mathbf{a} = \langle a_x, a_y \rangle$$

$$x_f = x_i + v_{x,i}t + \frac{1}{2}a_x t^2$$

$$y_f = y_i + v_{y,i}t + \frac{1}{2}a_y t^2$$



Clicker Question 4-1

For this fountain, what is the best guess for the acceleration ($\mathbf{a} = ??$) experienced by a fluid particle?

Assume y is positive upward; x is positive to the right.

1. $a_x \neq 0, a_y = g$
2. $a_x = 0, a_y = g$
3. $a_x \neq 0, a_y = -g$
4. $a_x = 0, a_y = -g$
5. Something else

Clicker Question 4-2

The average velocity for a macroscopic time step $\Delta t = t_f - t_i$ is given by:

$$\mathbf{v}_{avg} = \frac{\Delta \mathbf{r}}{\Delta t}$$

where $\Delta \mathbf{r} = \mathbf{r}_f - \mathbf{r}_i$. At what time do we estimate the average velocity occurs?

1. t_i
2. t_f
3. Sometime between t_f and t_i
4. $\frac{t_f - t_i}{2}$

Clicker Question 4-3

Consider the generic position vector $\vec{R}$ for a particle in 2D space. Which of the following describes the direction of the vector in plane polar coordinates (r, ϕ) ?

1. $\hat{R}$
2. $\hat{r}$
3. $\hat{\phi}$
4. Some combination of $\hat{r}$ and $\hat{\phi}$
5. I'm not sure.

Group Discussion 4-1

We found the following expression for the equation of motion of a falling ball subject to air resistance:

$$m\ddot{y} = +mg - b\dot{y} - c\dot{y}^2$$

What are the units of the constants b and c ?